

Lecture # 09

MM 318: Multicomponent Alloys

Dr. Sumanta Samal

Contact Details

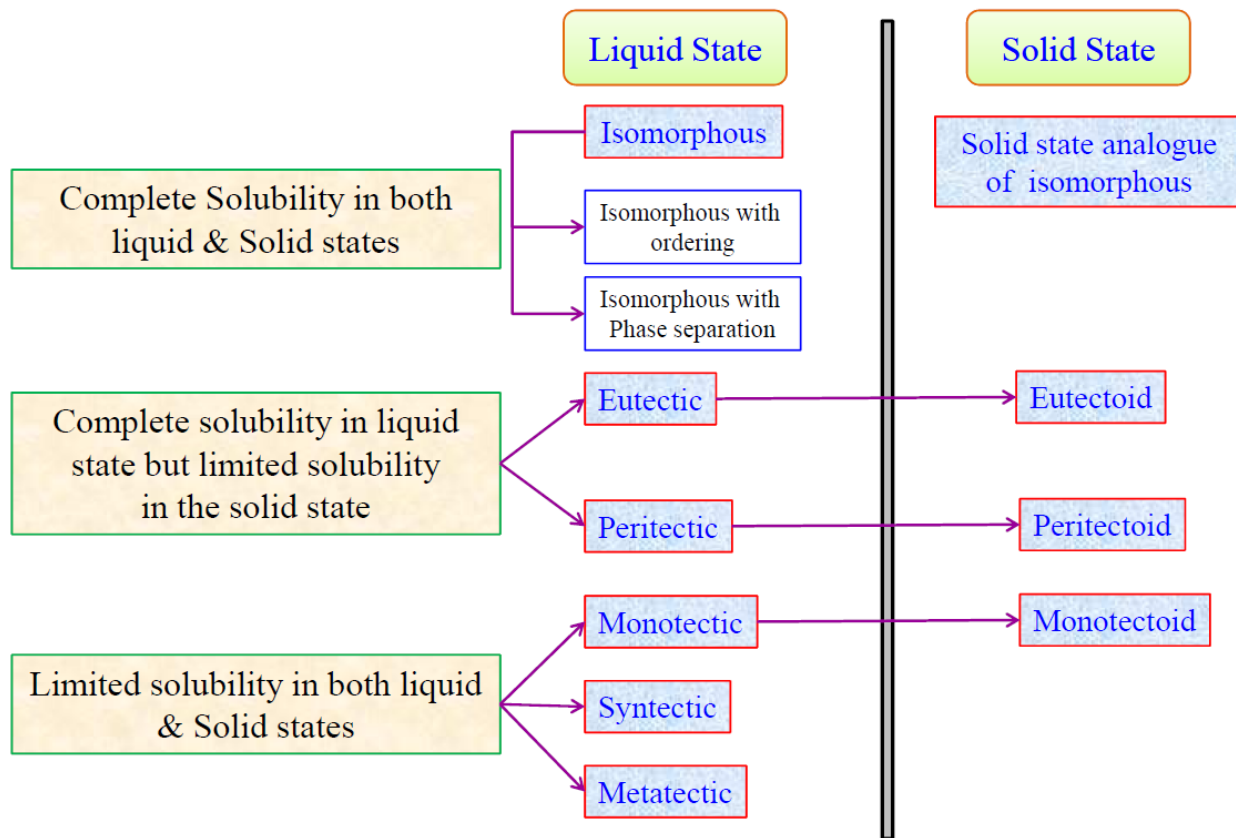
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Phase transformation



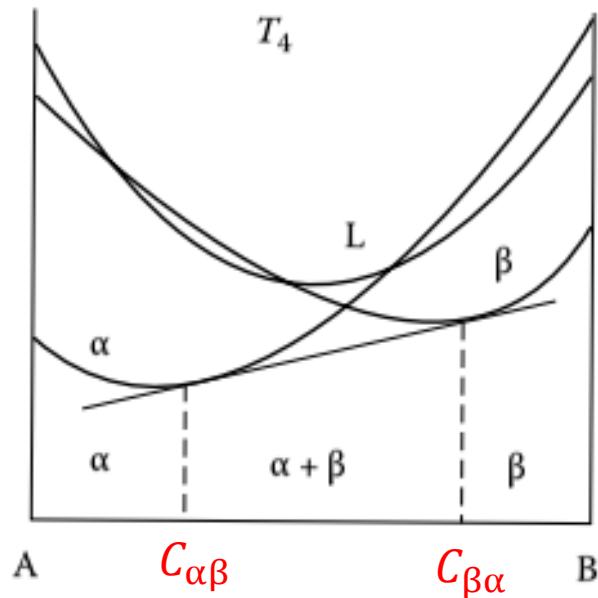
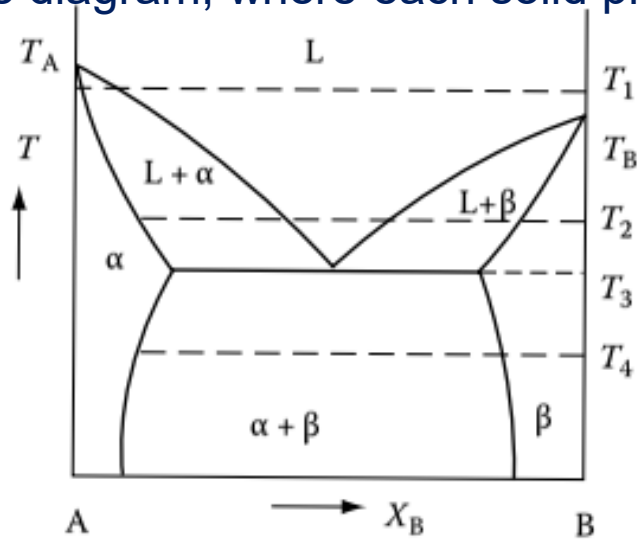
Name of reaction	Phase equilibrium	Schematic representation
Eutectic	$L \leftrightarrow S_1 + S_2$	
Peritectic	$S_1 + L \leftrightarrow S_2$	
Monotectic	$L_1 \leftrightarrow S_1 + L_2$	
Eutectoid	$S_1 \leftrightarrow S_2 + S_3$	
Peritectoid	$S_1 + S_2 \leftrightarrow S_3$	
Monotectoid	$S_{1a} \leftrightarrow S_{1b} + S_2$	
Metatectic	$S_1 \leftrightarrow S_2 + L$	
Syntectic	$L_1 + L_2 \leftrightarrow S$	

Overview of possible Binary Phase diagram

Summary of Invariant reactions

Eutectic growth

Eutectic Phase diagram, where each solid phase has a different crystal structure

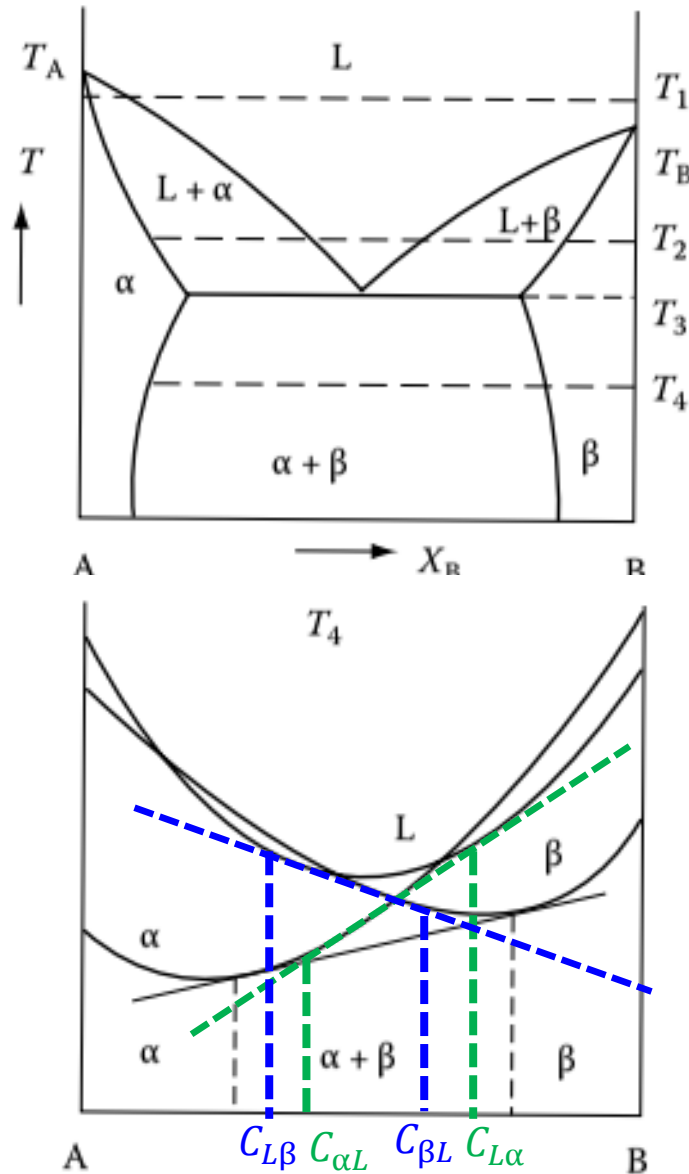


Eutectic reaction:
 $L \rightarrow \alpha + \beta$

$C_{\alpha\beta}$: Composition of α in equilibrium with β

$C_{\beta\alpha}$: Composition of β in equilibrium with α

Eutectic Growth



$C_{\alpha L}$: Composition of α in equilibrium with L

$C_{L\alpha}$: Composition of L in equilibrium with α

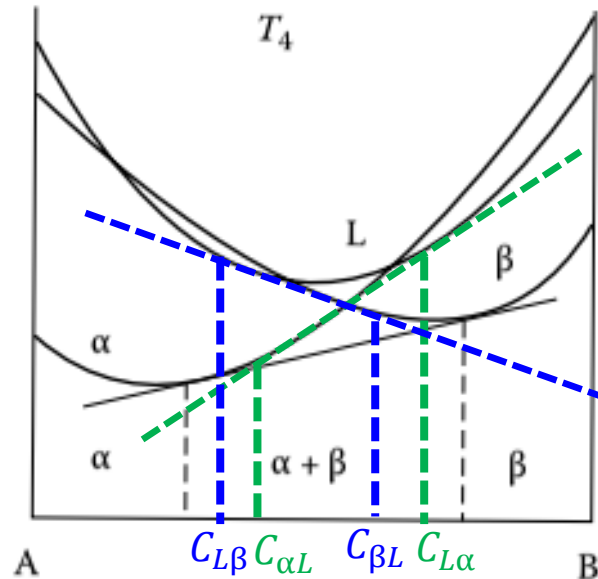
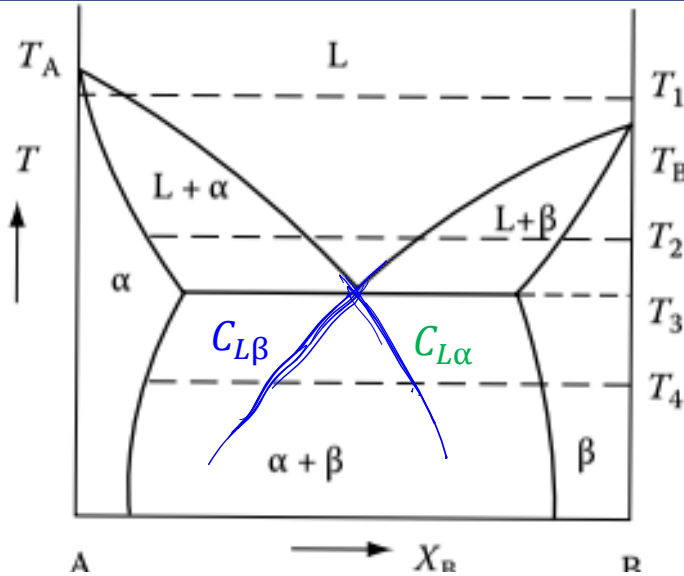
$C_{\beta L}$: Composition of β in equilibrium with L

$C_{L\beta}$: Composition of L in equilibrium with β

% in B content :

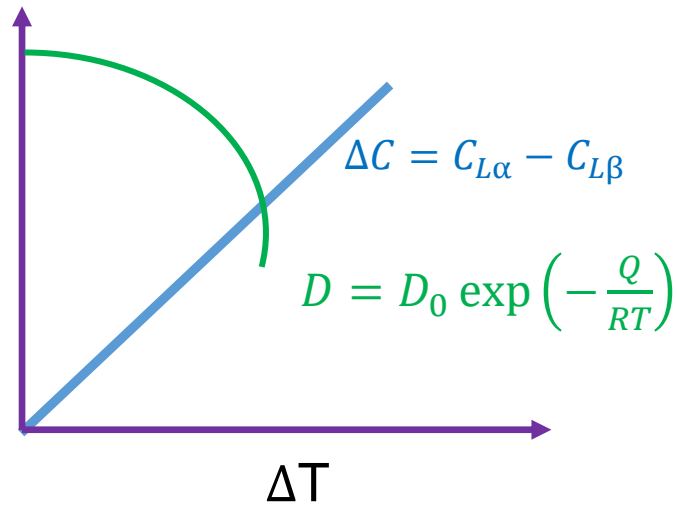
$$C_{L\alpha} > C_{L\beta}$$

Eutectic Growth



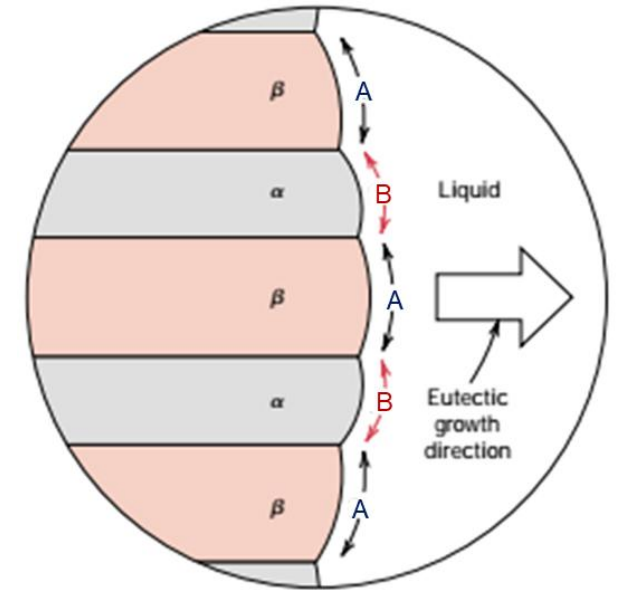
Eutectic reaction:
 $L \rightarrow \alpha + \beta$

% B atoms (solute) D



$C_{L\alpha}$: contains more B atoms (or less A atoms)

$C_{L\beta}$: contains less B atoms (or more A atoms)



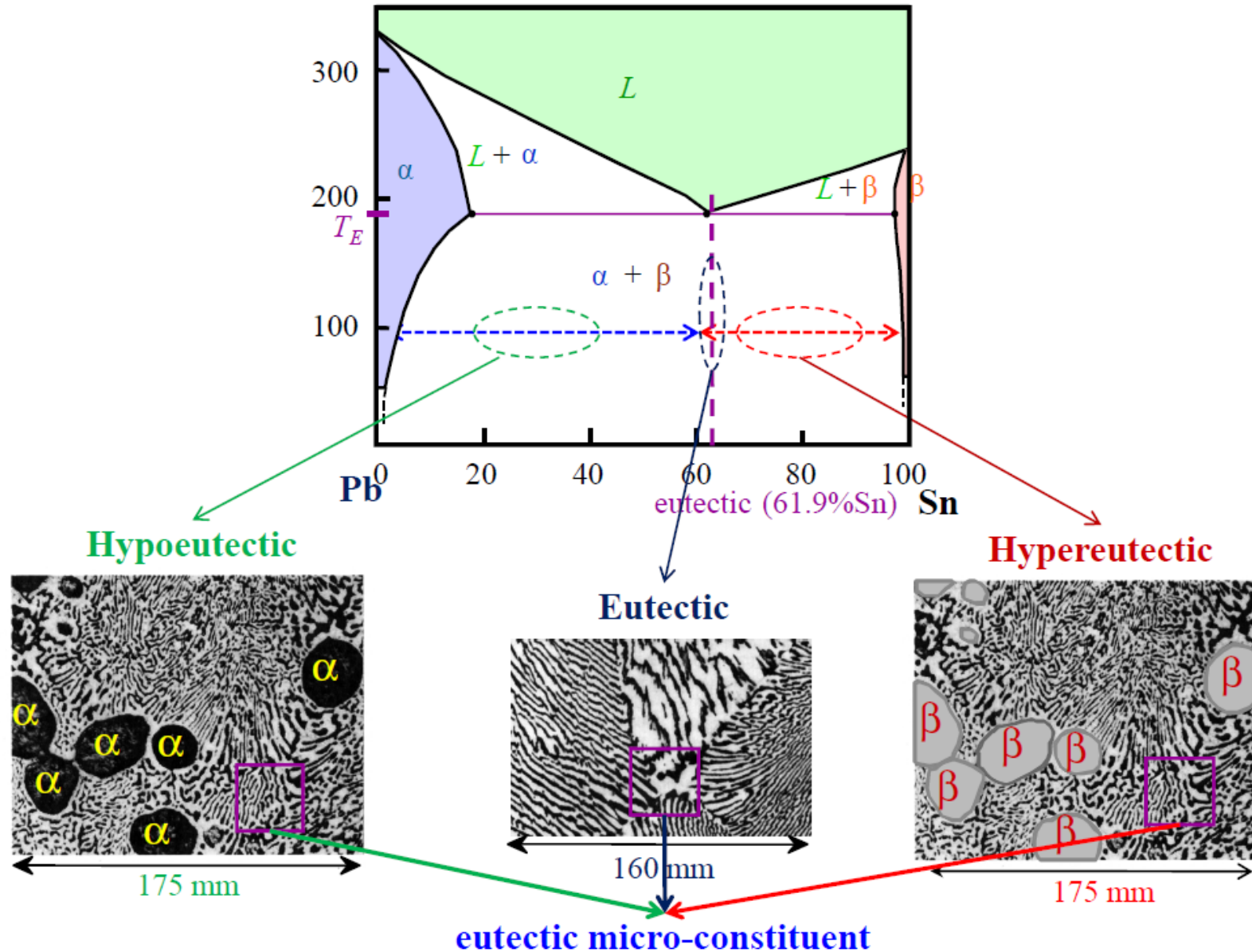
Eutectic growth mechanism

Fick's 1st Law: $J = -D \frac{\partial C}{\partial x}$

Three interfacial energy: $\gamma_{\alpha\beta}$, $\gamma_{\alpha L}$, $\gamma_{\beta L}$

- Driving force: $\Delta C = C_{L\alpha} - C_{L\beta}$
- For a particular temperature, ΔC is fixed
- With the increase in undercooling (ΔT), driving force (ΔC) increases
- With the increase in undercooling (ΔT), diffusivity (D) decreases

Hypo Eutectic – Eutectic – Hyper Eutectic



Peritectic Growth

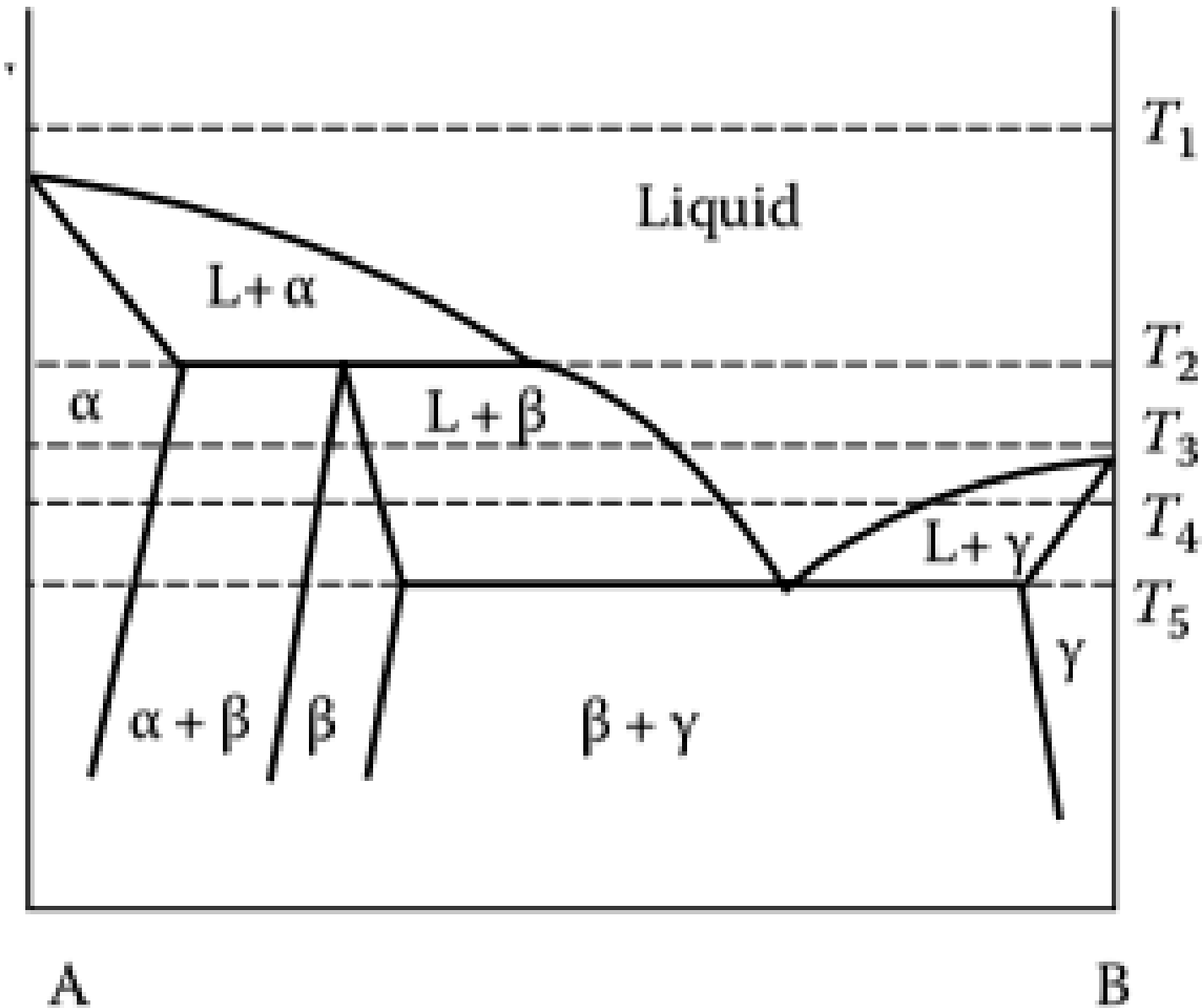
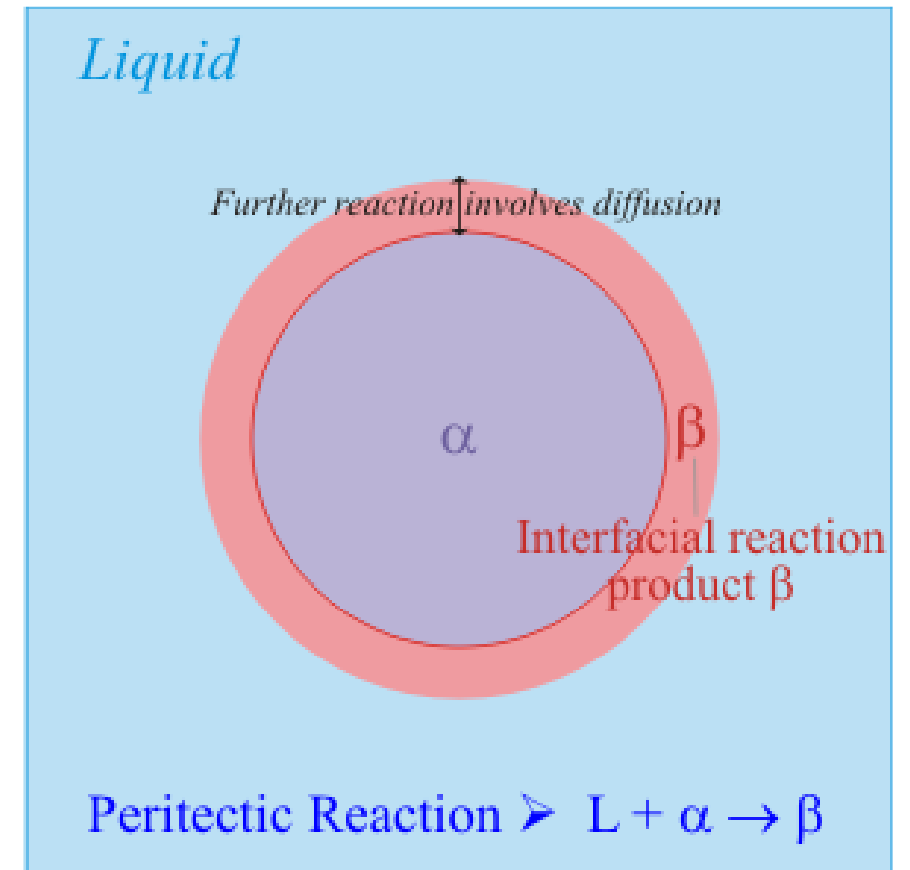
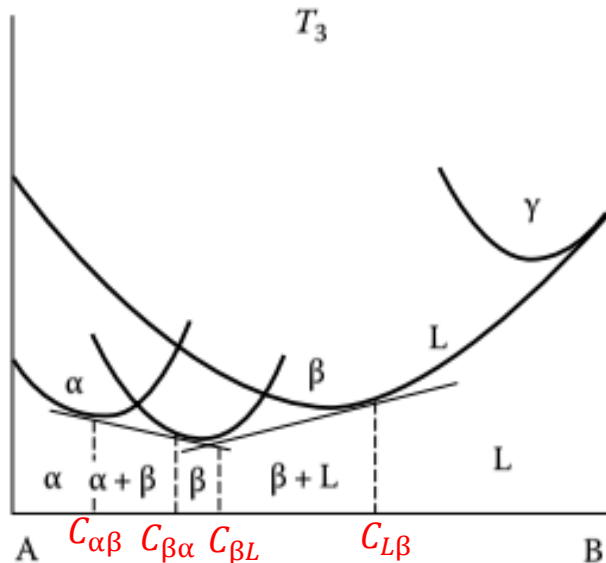
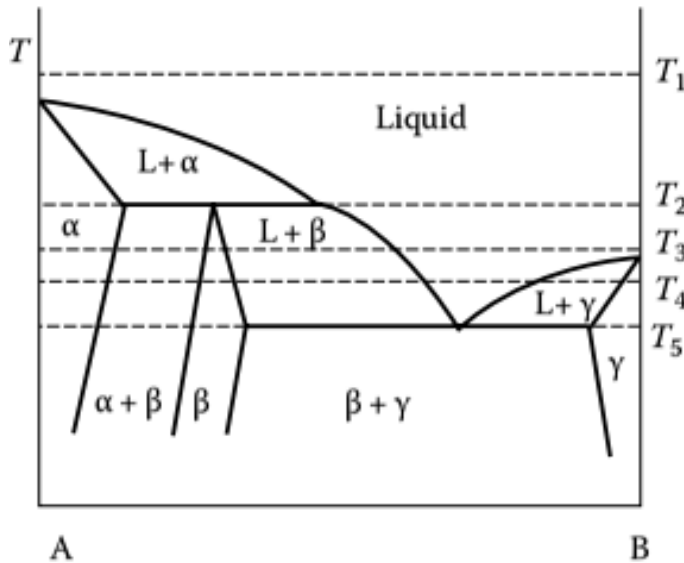


Fig. The derivation of a complex phase diagram.



Peritectic Growth



$C_{\alpha\beta}$: Composition of α in equilibrium with β

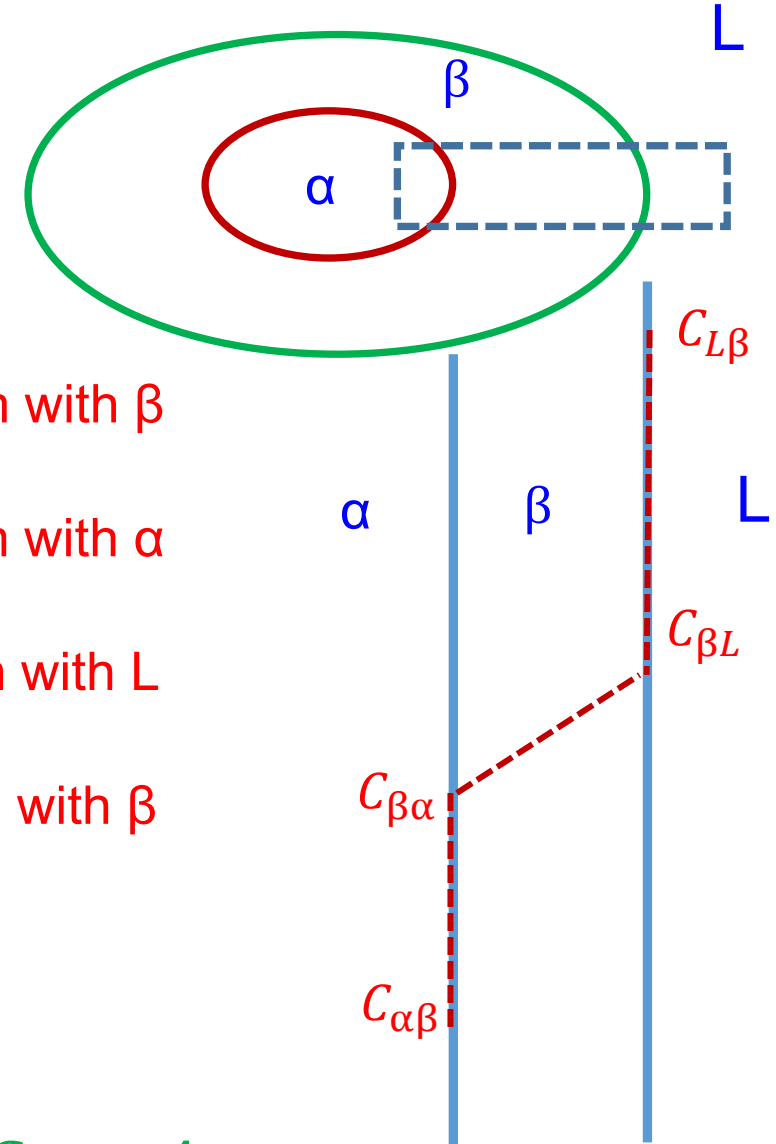
$C_{\beta\alpha}$: Composition of β in equilibrium with α

$C_{\beta L}$: Composition of β in equilibrium with L

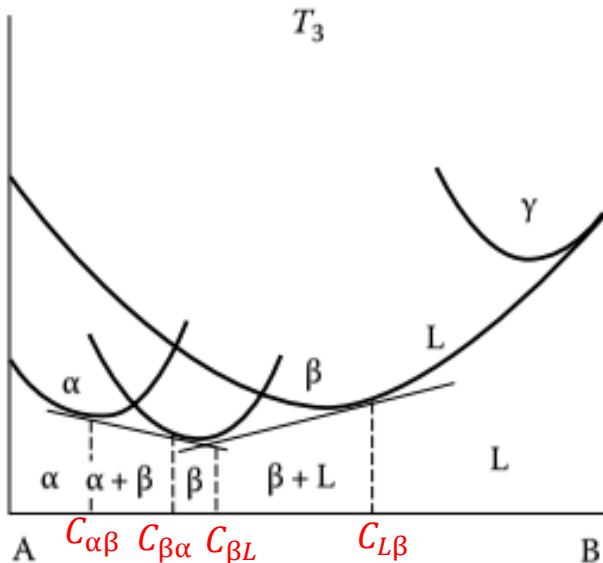
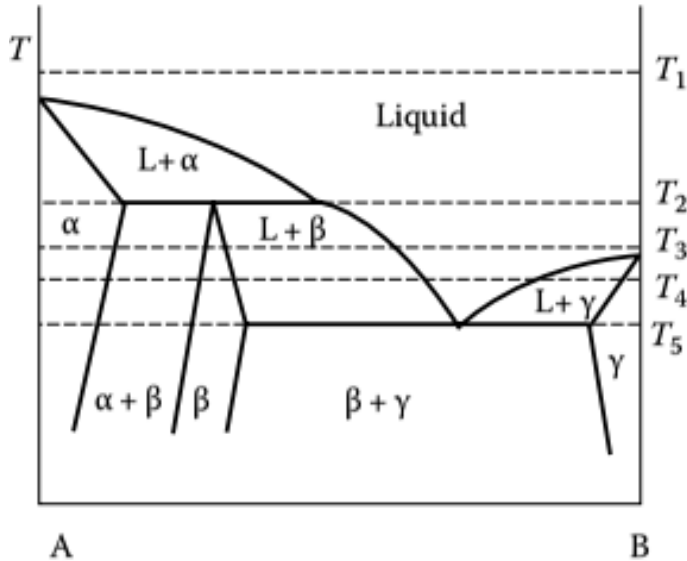
$C_{L\beta}$: Composition of L in equilibrium with β

Nucleation:

$$\text{Change in energy } (\Delta G) = V\Delta G_V + A\gamma$$



Peritectic Growth



Peritectic reaction:
 $L + \alpha \rightarrow \beta$

Fick's 1st Law: $J = -D \frac{\partial c}{\partial x}$

→ Indicates flow of excess % B (solutes)

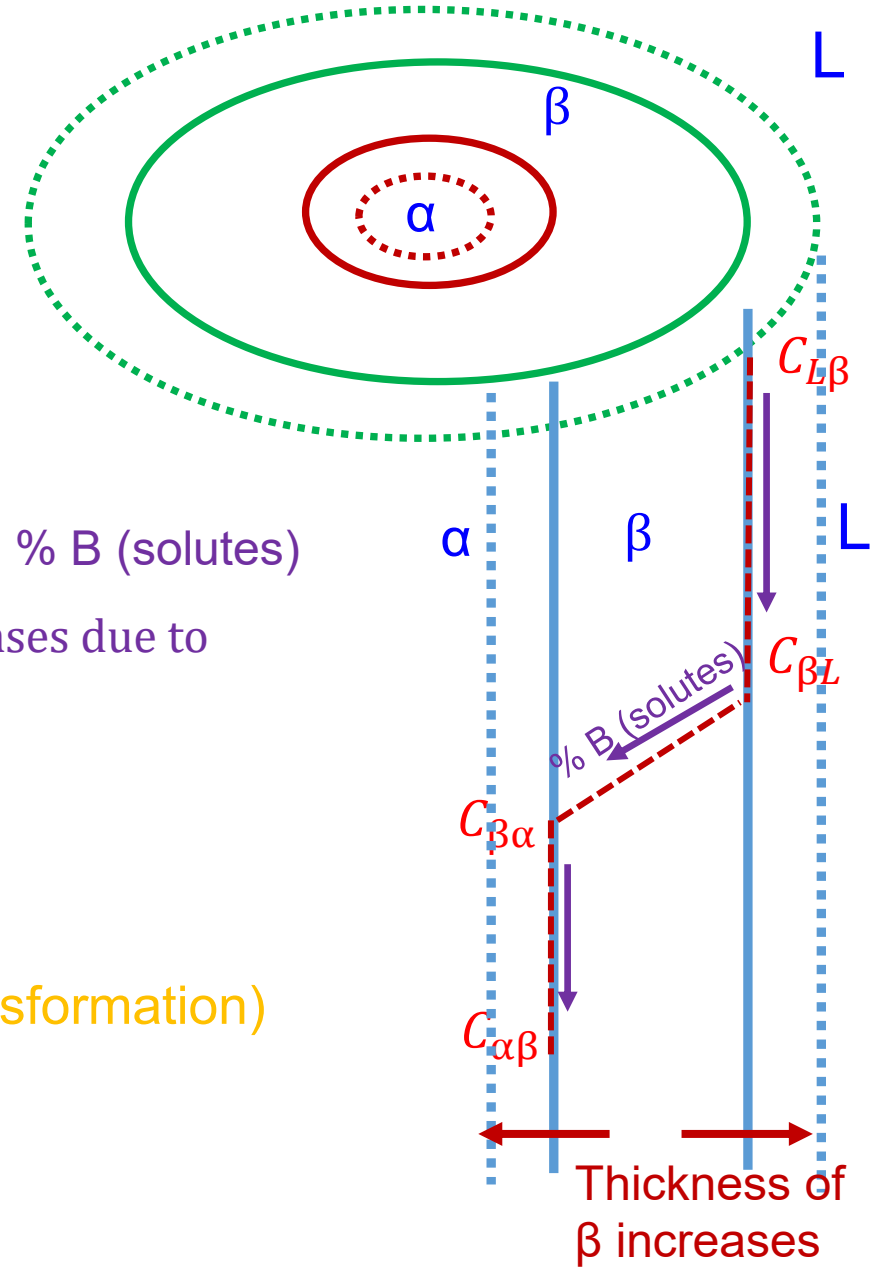
➤ the thickness of β phase increases due to

Peritectic transformation:

1. $L \rightarrow \beta + B(\text{solute})$ (faster)

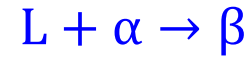
2. $\alpha + B(\text{solute}) \rightarrow \beta$

(slower due to solid state transformation)



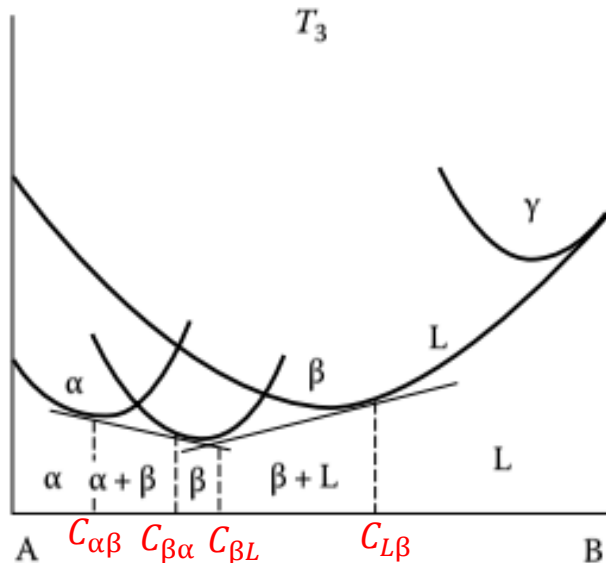
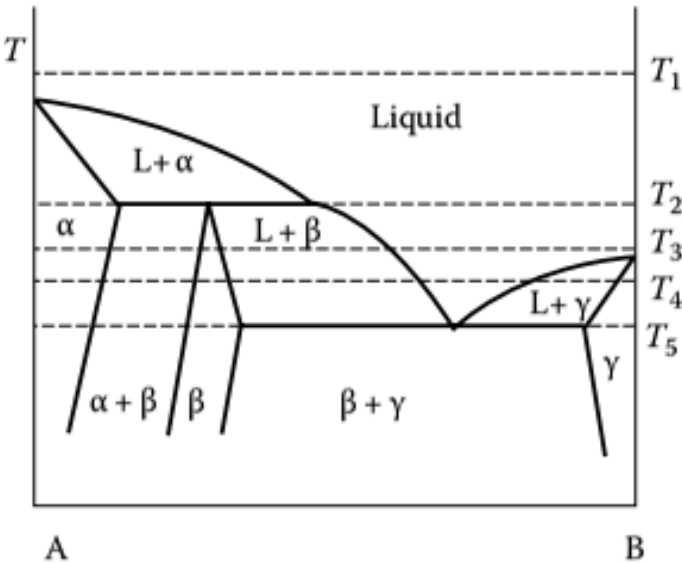
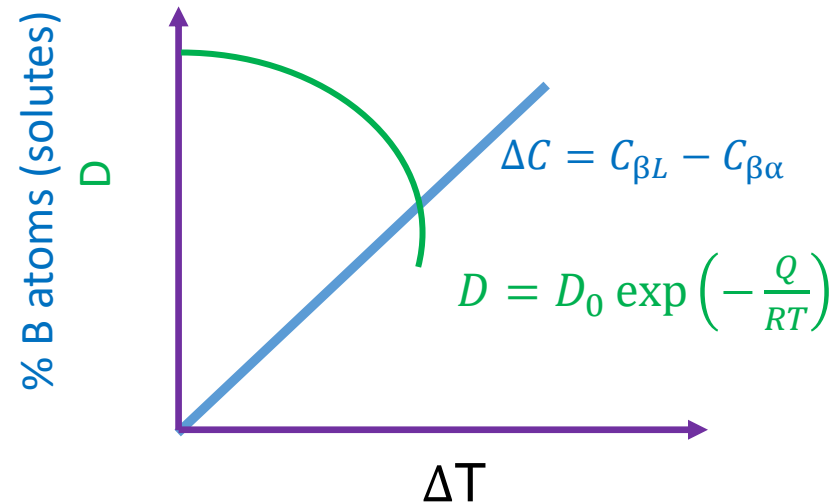
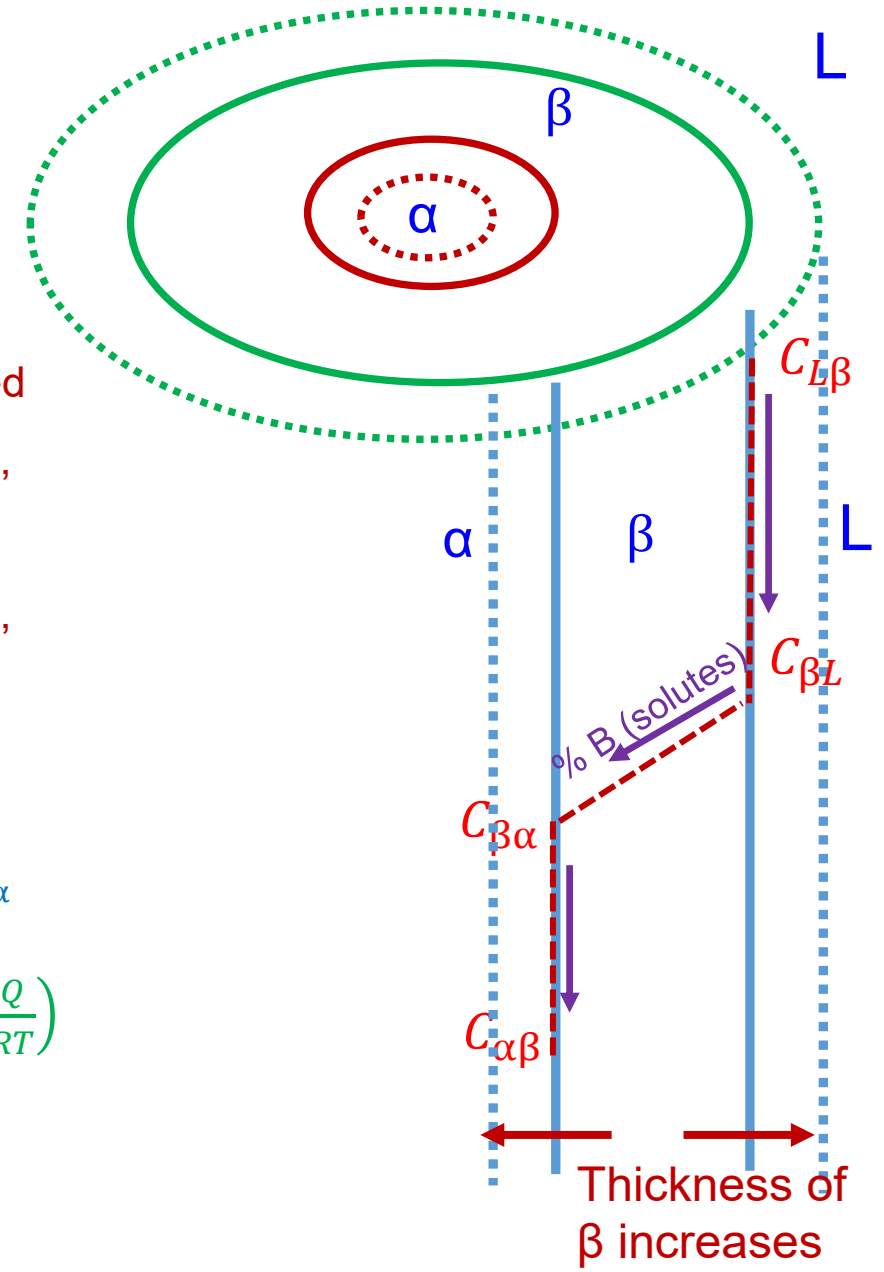
Peritectic Growth

Peritectic reaction:

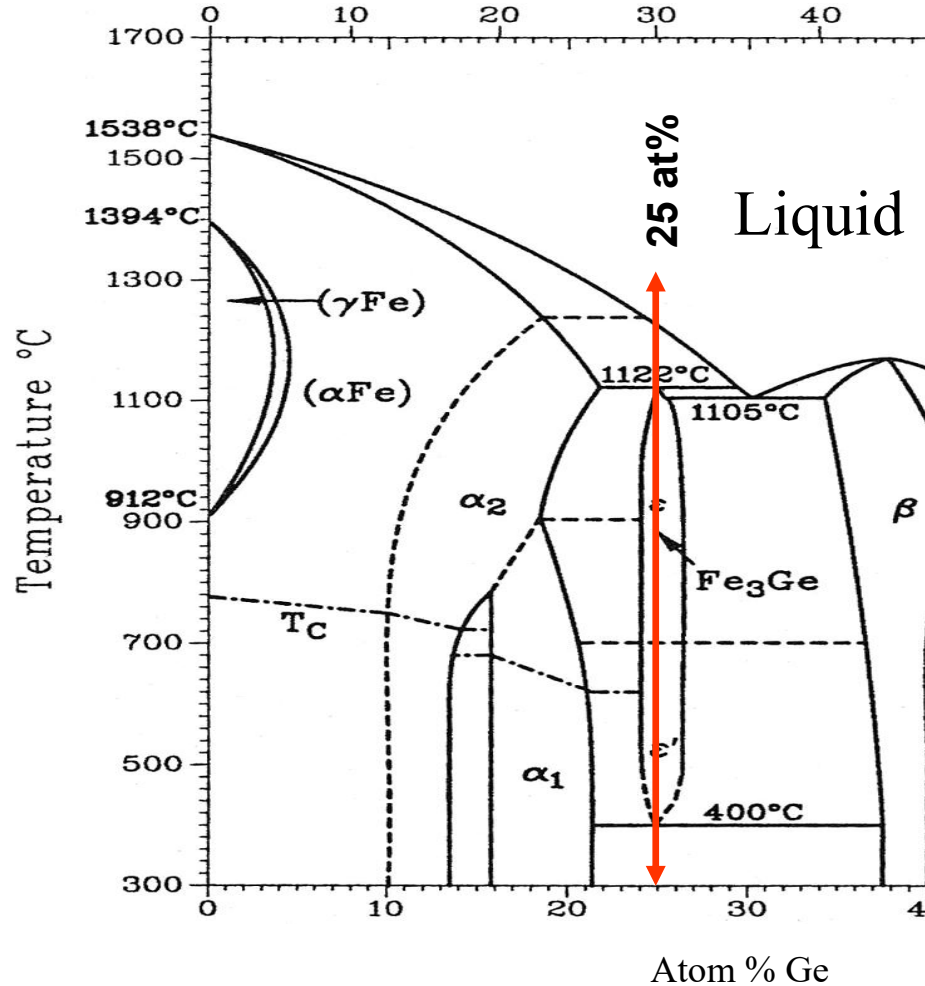


Fick's 1st Law: $J = -D \frac{\partial C}{\partial x}$

- Driving force: $\Delta C = C_{\beta L} - C_{\beta \alpha}$
- For a particular temperature, ΔC is fixed
- With the increase in undercooling (ΔT), driving force (ΔC) increases
- With the increase in undercooling (ΔT), diffusivity (D) decreases



Peritectic growth



Peritectic reaction: $L + \alpha \rightarrow \epsilon$

Followed by Eutectic reaction: $L \rightarrow \beta + \epsilon$

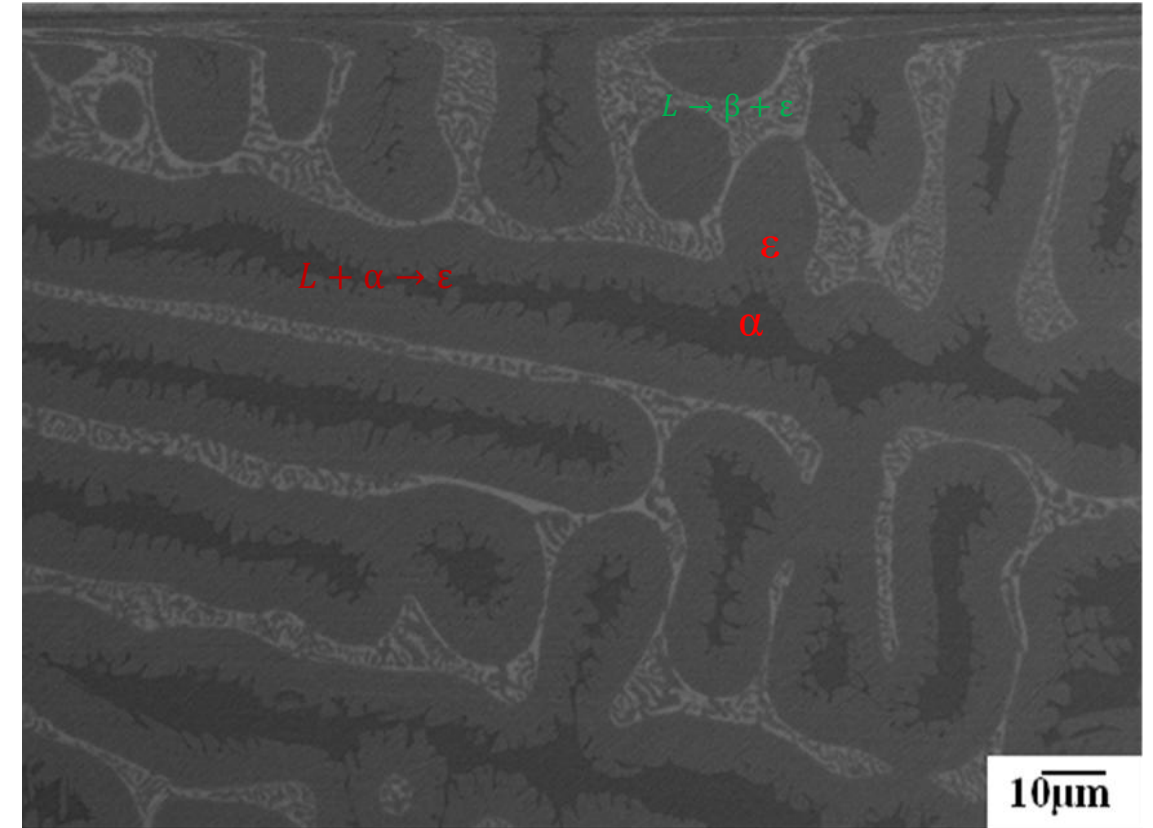
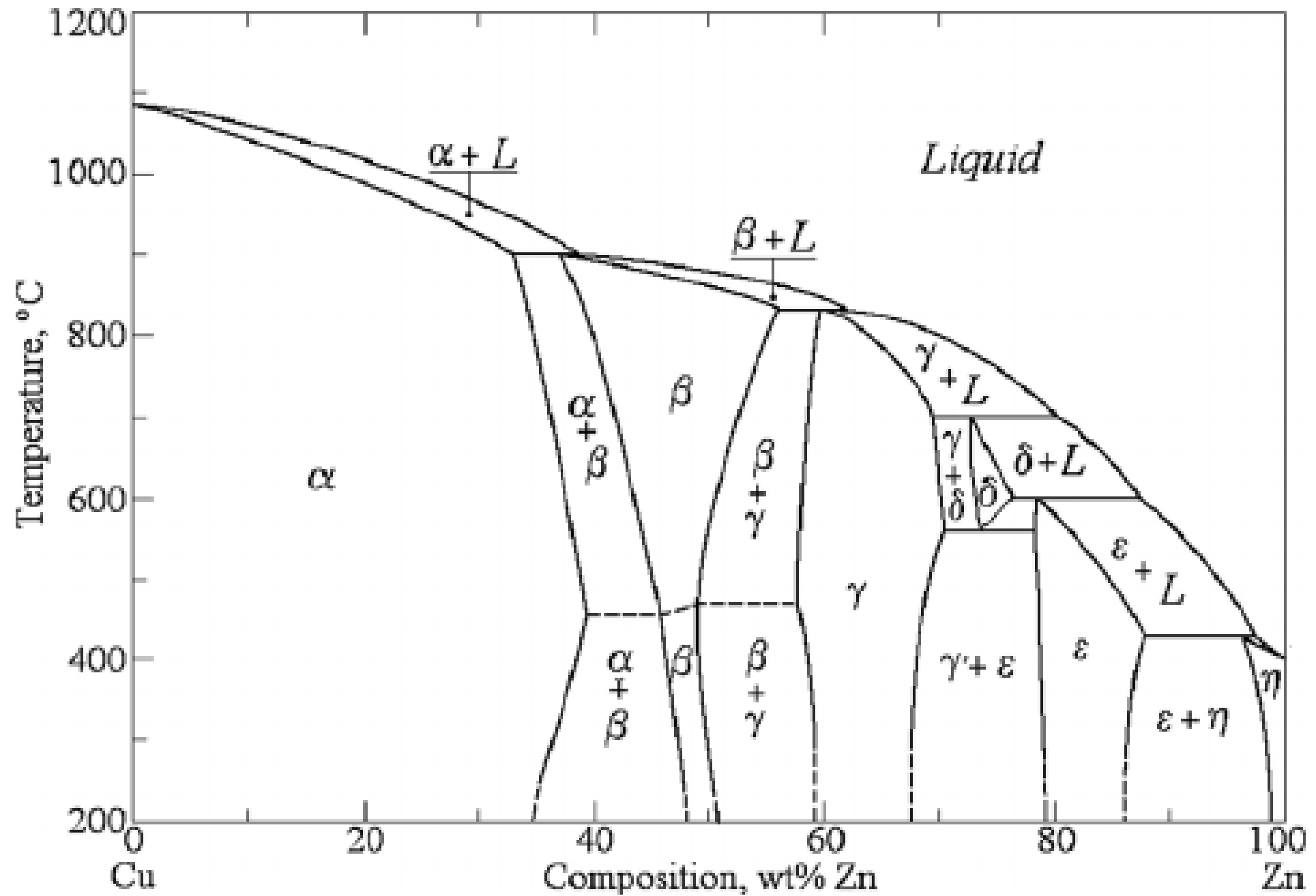


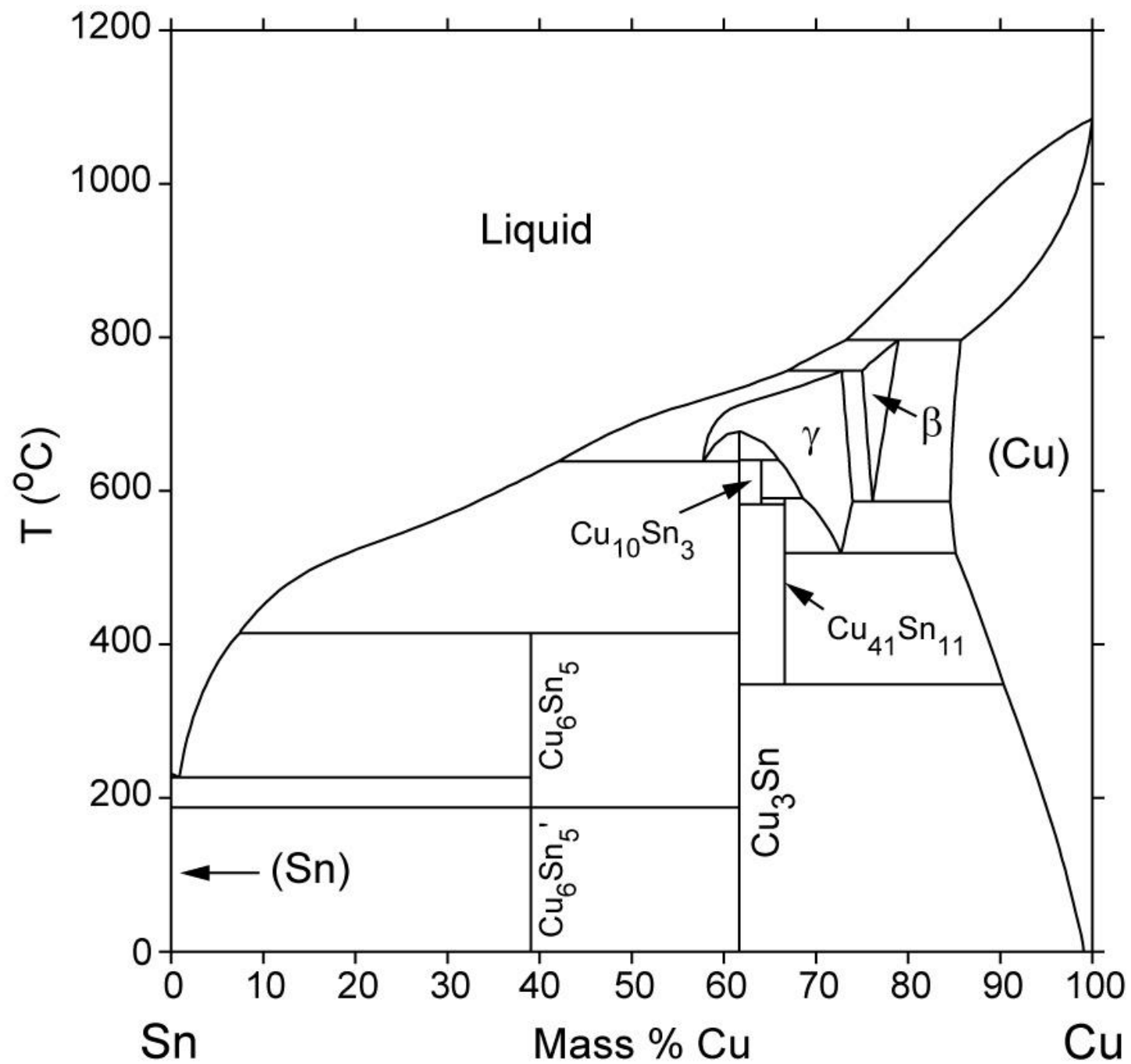
Figure: As-cast Microstructures of Fe-25 atom% Ge alloy

dark phase is α , gray phase is ϵ and white phase is β .

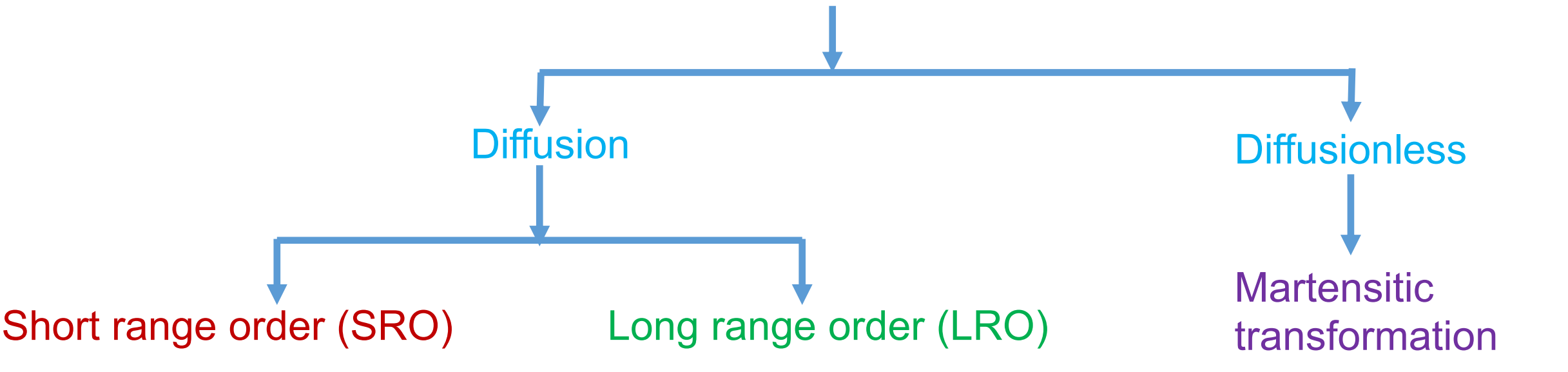
Brass



Bronze



Solid state transformation



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graph TD; A[Solid state transformation] --> B[Diffusion]; A --> C[Diffusionless]; B --> D[Short range order (SRO)]; B --> E[Long range order (LRO)]; C --> F[Martensitic transformation];
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Diffusion

Diffusionless

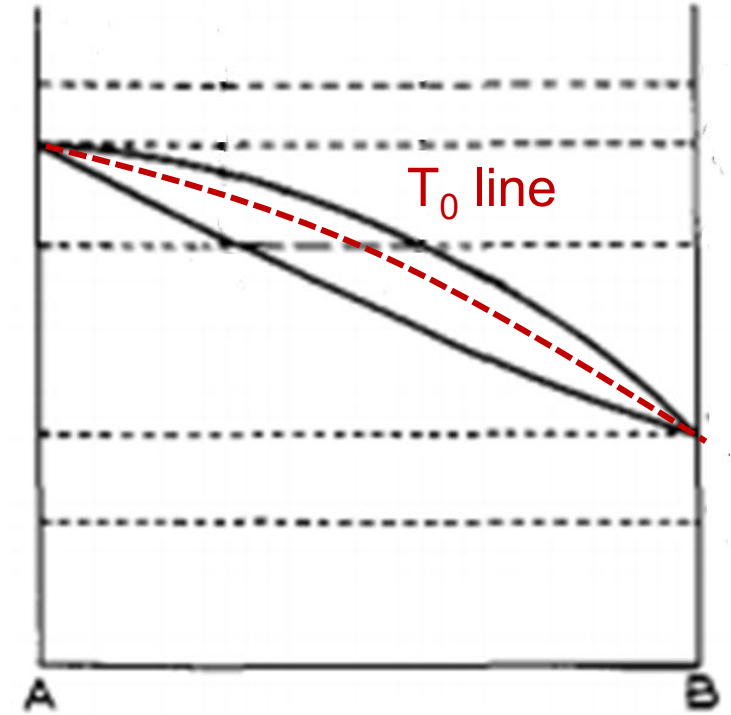
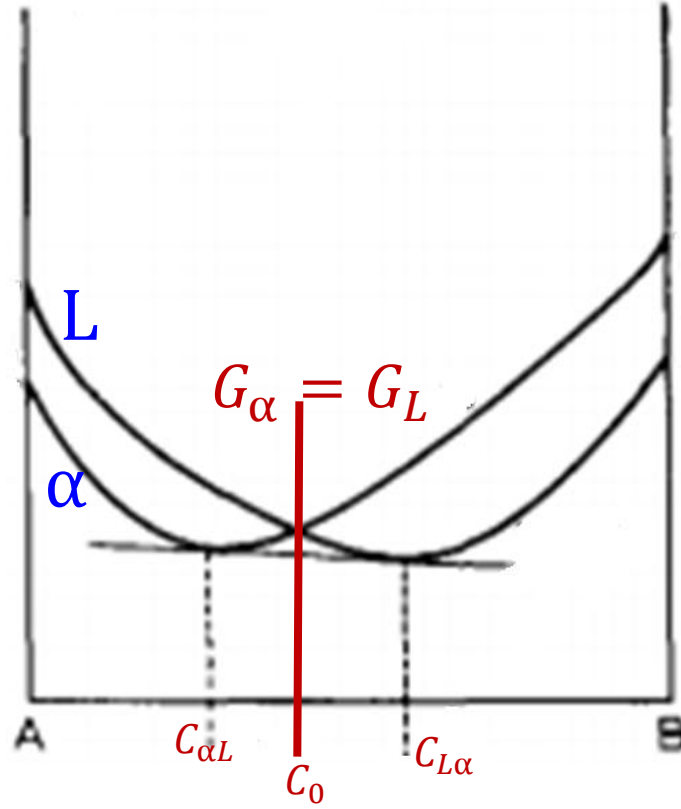
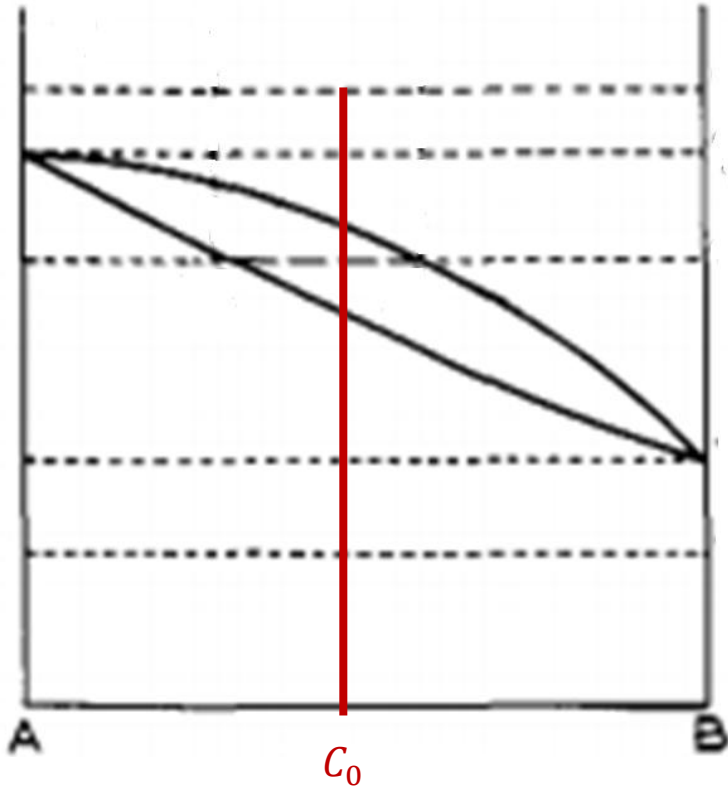
Short range order (SRO)

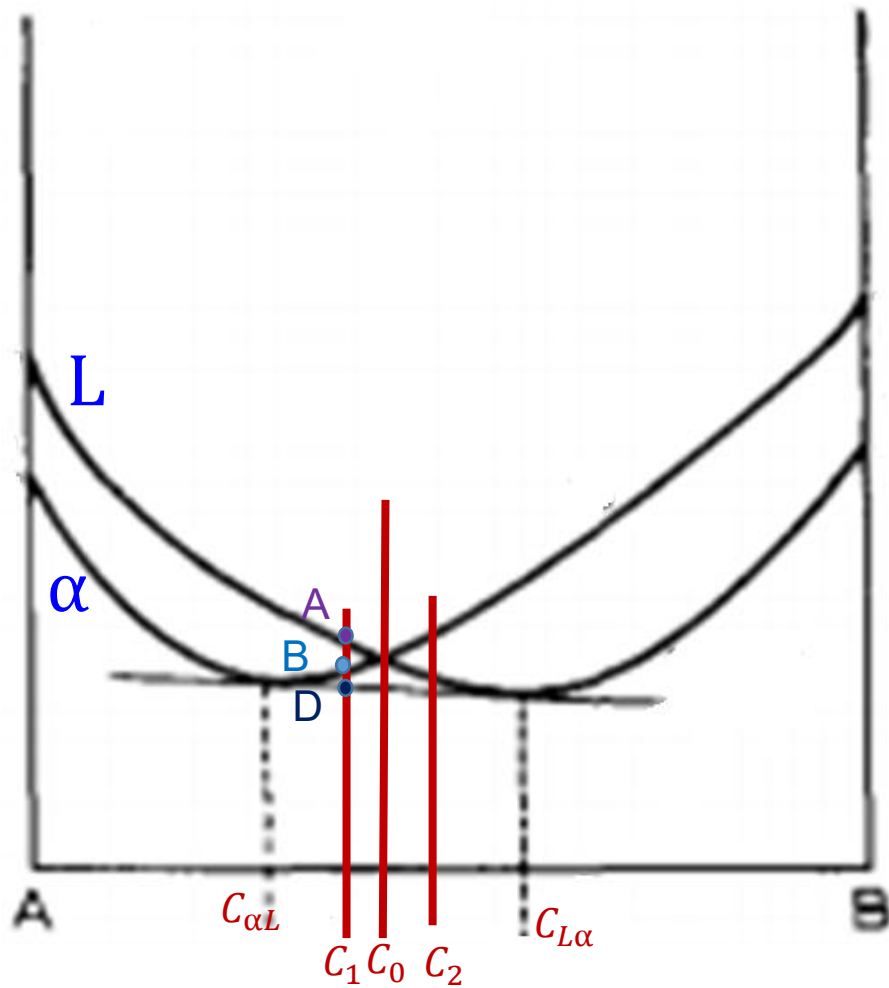
Long range order (LRO)

Martensitic transformation

- Allotropic
- Massive
- Spinodal decomposition
- Order-disorder transformation
- Eutectoid (Pearlite, Bainite transformation etc.)
- Peritectoid
- Precipitation

T_0 Concept





A : Composition of liquid having composition $C_1 : (L)_{C_1}$

B : Composition of solid (α) having composition $C_1 : (\alpha)_{C_1}$

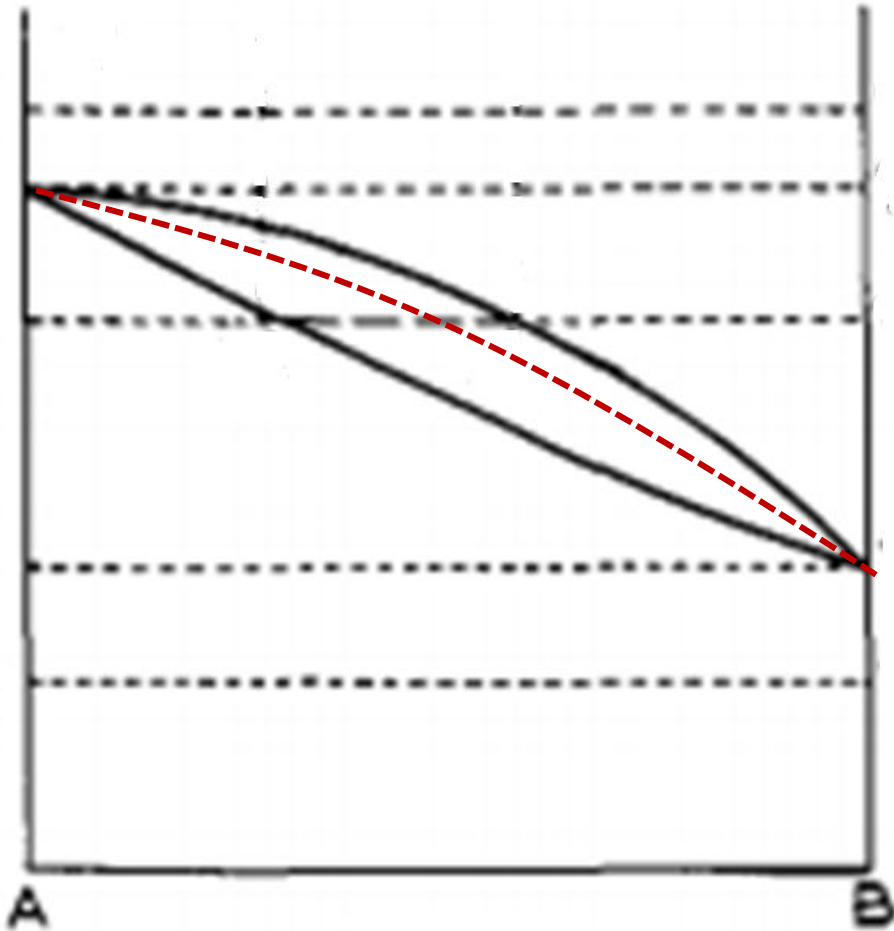
(Partitionless solidification or Polymorphous transformation)

D : Composition at point D having two phase mixture: $(\alpha)_{C_{\alpha L}} + (L)_{C_{L\alpha}}$

At C_2 composition; Gibbs free energy of α phase having C_2 composition is greater than that of Gibbs free energy of L phase having C_2 composition

No Partitionless solidification

T_0 Curve



- If nucleation temperature is below T_0 (i.e. undercooling is very large); there is possibility of having **Partitionless Solidification**
- If nucleation temperature is above T_0
- (i.e. undercooling is low);

No Partitionless Solidification

T_0 concept

- Formation of Primary (α) phase only (very high undercooling)
- Nucleation of secondary (β) bypassing α (very fast cooling rate)

Thank you